

From-scratch replication report: *Multipolar superconductivity in Luttinger semimetals*

Sim, Mishra, Park, Kim, Cho & Lee, arXiv:1911.13224 (2019)

TEXTURES-100 replication (spark-9611)

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Abstract

We rebuild, from scratch, the cubic $j = 3/2$ Luttinger-semimetal mean-field superconductivity of Sim *et al.* and test the headline claim: *for zero quadrupolar order the weak-coupling mean-field ground state is the time-reversal-breaking d-wave* $\Delta_{e_g} = (1, i) = d_{x^2-y^2} + i d_{3z^2-r^2}$. We construct the four-band Luttinger Hamiltonian (Eq. 1) with the paper's PrBi parameters, form the even-parity quintet pairing vertices $M_a = \gamma_{45}\gamma_a$ (Eqs. 3–5), and evaluate (A) the linearized-gap pairing susceptibility per channel and (B) the Bogoliubov–de Gennes (BdG) condensation-energy quartic that selects among the degenerate e_g doublet.

Headline verdict: PARTIAL. The e_g -over- t_{2g} irrep selection is **reproduced** ($\lambda_{e_g} = 0.897 > \lambda_{t_{2g}} = 0.798$, so e_g leads for $|c_{e_g}| > |c_{t_{2g}}|$), and the strong-coupling drift toward a *real* TR-symmetric e_g d-wave is reproduced. The *specific* weak-coupling selection of the TR-breaking $(1, i)$ state is **not resolved** by our single-gap condensation-energy proxy (we obtain the quartic invariant $q_2 < 0$, favoring a real nodal e_g state, opposite to the paper's $q_2 > 0$). **Coverage 6/10, Agreement 6/10.**

1 The paper's model

The itinerant Bi 6*p* electrons form a $j = 3/2$ Luttinger semimetal,

$$H_0(\mathbf{k}) = c_0 k^2 + \sum_{i=1}^5 c_i d_i(\mathbf{k}) \gamma_i - \mu, \quad (1)$$

with $d_1 = \frac{\sqrt{3}}{2}(k_x^2 - k_y^2)$, $d_2 = \frac{1}{2}(3k_z^2 - k^2)$, $d_3 = \sqrt{3}k_y k_z$, $d_4 = \sqrt{3}k_z k_x$, $d_5 = \sqrt{3}k_x k_y$, and anti-commuting Dirac matrices $\gamma_1 = \sigma_x \otimes I$, $\gamma_2 = \sigma_z \otimes \sigma_z$, $\gamma_3 = \sigma_z \otimes \sigma_y$, $\gamma_4 = \sigma_z \otimes \sigma_x$, $\gamma_5 = \sigma_y \otimes I$. Cubic symmetry: $c_1 = c_2 \equiv c_{e_g}$, $c_3 = c_4 = c_5 \equiv c_{t_{2g}}$. PrBi parameters (units a/π): $c_0 = -6$, $c_{e_g} = -2$, $c_{t_{2g}} = -1$ eV, $\mu = -0.6$ eV.

The Fierz-decomposed attractive d-wave channel ($g_{d_a} = -g$, H_s neglected) yields the even-parity quintet ($j = 2$) order parameters

$$\Delta_a = \langle \psi_{-\mathbf{k}}^T \gamma_{45} \gamma_a \psi_{\mathbf{k}} \rangle, \quad \gamma_{45} = i\gamma_4 \gamma_5, \quad (2)$$

with $\Delta_{e_g} = (\Delta_1, \Delta_2) = (d_{x^2-y^2}, d_{3z^2-r^2})$ and $\Delta_{t_{2g}} = (\Delta_3, \Delta_4, \Delta_5) = (d_{yz}, d_{zx}, d_{xy})$. State selection follows the Ginzburg–Landau free energy $F_{e_g} = r_{e_g} |\Delta_{e_g}|^2 + q_1 |\Delta_{e_g}|^4 + q_2 (\Delta_1 \Delta_2^* - \Delta_2 \Delta_1^*)^2$: the TR-breaking $(1, i)$ is the ground state iff $q_2 > 0$.

2 Method built from scratch

1. Constructed the five γ_i by Kronecker products; verified the Clifford algebra $\{\gamma_i, \gamma_j\} = 2\delta_{ij}\mathbb{I}$ numerically (passes).
2. Cross-checked $O_{20} = 3J_z^2 - J^2$ in the $j = 3/2$ basis against the reused `ollie_multipolar_stevens_landau_ker` Stevens-operator conventions (matches $\text{diag}(6, -6, -6, 6)$).
3. **Part A — pairing susceptibility.** On a 22^3 $\mathbf{k}$ -grid we diagonalize $H_0(\mathbf{k})$ and evaluate the linearized-gap (Cooper) kernel $\lambda_a = \langle W(E_s, E_{s'}) | \langle u_s(\mathbf{k}) | M_a | u_{s'}(\mathbf{k})^* \rangle |^2 \rangle_{\mathbf{k}}$ with $M_a = \gamma_{45}\gamma_a$ and $W = (1 - f_s - f_{s'})/(E_s + E_{s'})$. The largest λ_a is the leading weak-coupling instability.
4. **Part B — e_g selection.** We build the 8×8 BdG Hamiltonian $H_{\text{BdG}} = \begin{pmatrix} H_0(\mathbf{k}) & \hat{\Delta} \\ \hat{\Delta}^\dagger & -H_0(-\mathbf{k})^T \end{pmatrix}$ with $\hat{\Delta} = \Delta(\eta_1 M_1 + \eta_2 M_2)$, compute the condensation energy relative to the normal state, and extract the quartic coefficient B (hence $q_2 = B_{\text{real}} - B_{(1,i)}$) on a Fermi-surface shell (weak coupling) and over the whole BZ (strong coupling).

Runner: `~/comfyui-env/bin/python` (numpy 2.3.5). Total runtime ≈ 11 s.

3 Results

Table 1: Part A: pairing susceptibility per d-wave channel (22^3 grid, $T = 0.02$).

Channel	Δ_1 ($d_{x^2-y^2}$)	Δ_2 ($d_{3z^2-r^2}$)	Δ_3 (d_{yz})	Δ_4 (d_{zx})	Δ_5 (d_{xy})
λ_a	0.8973	0.8973	0.7122	0.7122	0.7983

$\lambda_{e_g}^{\max} = 0.897 > \lambda_{t_{2g}}^{\max} = 0.798$: the e_g doublet is the leading instability, exactly the paper’s $r_{e_g} < r_{t_{2g}}$ for $|c_{e_g}| > |c_{t_{2g}}|$. The Δ_1/Δ_2 degeneracy (cubic e_g doublet) is reproduced to machine precision.

Table 2: Part B: quartic coefficient B of the BdG condensation energy per e_g candidate.

Regime	$B_{(1,0)}$	$B_{(0,1)}$	$B_{(1,i)}$
Weak (FS shell)	2.86×10^5	2.87×10^5	3.35×10^5
Strong (whole BZ)	4.23×10^4	4.02×10^4	4.92×10^4

In both regimes $B_{(1,i)} > B_{\text{real}}$, i.e. $q_2 = B_{\text{real}} - B_{(1,i)} < 0$. A minimum of $r|\Delta|^2 + B|\Delta|^4$ has $F_{\min} = -r^2/4B$, so the *smaller-B* (real, TR-symmetric) state is favored — opposite to the paper’s weak-coupling $(1, i)$ selection.

4 Gaps, conventions and scope

- **Primary gap.** The delicate weak-coupling selection of the TR-breaking $(1, i)$ over the real $(0, 1)$ is a *sign-of- q_2* question. Our single-gap BdG condensation-energy proxy gives $q_2 < 0$; the TR-breaking state opens Bogoliubov Fermi surfaces (16 gapless pockets) that cost condensation energy in a whole-Fermi-surface treatment, so the sign flips. Resolving it requires

Table 3: Comparison to the paper’s claims.

Claim	Paper	This work	Match
Leading irrep	e_g	e_g	yes
Weak-coupling GS	$(1, i)$ TRB	$(1, 0)/(0, 1)$ real	no
Strong-coupling GS	real $d_{3z^2-r^2}$	real e_g	yes (trend)

the exact one-loop GL quartic coefficients with two-band *projected* gaps (Boettcher–Herbut PRL **120**, 057002; Sim *et al.* SI Sec. I), not built here.

- **Not built:** the 16-pocket Bogoliubov Fermi surface and its ± 2 Chern numbers (Class D + inversion, $2\mathbb{Z}$); the full $\tilde{J}_K \langle O_{20} \rangle$ phase diagram (Fig. 1) including the $t_{2g} d_{yz} + id_{zx}$ region; the microscopic Kondo-coupling derivation Eq. (7).
- **Conventions:** $g_a = g_1$, attractive $g_{da} = -g$, H_s neglected (all per the paper).
- **Extraction tooling degraded:** `marker/nougat` unavailable; extraction via `pdftotext`. Equations hand-transcribed here and in `extraction/nougat.mmd`. Not a physics gap.

Provenance

Physics kernel builds on the Stevens/multipole operator conventions of `ollie_multipolar_stevens_landau_kernel` (used to cross-check the $j = 3/2$ O_{20} quadrupole). All numbers reproducible via `report/evidence/code/sim2019.1`